MICROBIOLOGY

CHAPTER24

Other Gram Positive
Bacteria: Actinomycetes

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CH24: Other Gram Positive Bacteria: Actinomycetes

In this chapter we'll study members of class Actinobacteria (Many Actinobacteria are also called Actinomycetes) :

- Their life cycle
- cell wall types
- cell sugar patterns
- characteristics used in actinomycete taxonomy
- Their ecological significance
- Different groups in Actinomycetes

- Actinomycetes
  - The main difference between Actinomycetes and fungi is that it is a prokaryote that can't be seen with naked eye. As for fungi it can be seen since it is a eukaryote
  - They are the source of most currently used antibiotics
  - also produce metabolites that are anticancer, anthelminthic* and immunosuppressive** drugs
  - their growth involves a complex life cycle
- Life Cycle of Actinomycetes
  - involves development of filamentous cells (**hyphae**) and Spores
  - hyphae can form branching network
    - can grow on surface of substrate or into it to produce a **substrate mycelium**
    - some hyphae differentiate to form an ***aerial* mycelium** which extends above substratum →form exospores which are called sporangiospores if they are located in a sporangium
    - at this stage forms secondary metabolites some of which are medically useful

* Anthelmintic drugs: drugs that expel parasitic worms (helminthes) and other internal parasites from the body by either stunning or killing them and without causing significant damage to the host.

**Immunosuppressive drugs: are used in transplant surgeries to prevent the immune system from reacting strongly until the body gets used to the new organ

شكل الactinomycetes بشبه الfungi. بكون عندها تركيبات بتشبه الخيوط منسميها **Hypha** ←مجموعة الhypha منسميها **hyphae** ← مجموعة م الhyphae مع بعض منسميهم **mycelium** ←مجموعة الmycelium بتعطي**mycelia** ←مجموعة الmycelia بتعطي **Thullus**

الhyphae بتنموا بشكل اساسي على شكل شبكة متفرعة بطريقيتين:

يا اما بتنمو على سطح substrate معين(متل الagar) او داخله على شكل substrate mycelium

او بتمتد برا الsubstrate على شكل aerial mycelium الexospores الي بشكلوهم منسميهم sporangiospores لأنهم محفوظين بكيس او sack منسميه sporangium.

- more about Actinomycetes
  - most are not motile
    → in those that are, motility is restricted to flagellated spores

معظم الactinomycetes ما بتحركوا ولكن الي منهم بتحركوا بتكون حركتهم بس مقتصرة على الspores الي بكون الها flagella

  - Spores: most are NOT heat resistant but withstand desiccation

بنتجوا exospores وتزكروا انها مو متل الendospores بمقاومتها للحرارة وما بتقدر تقاومها ولكنها بتقدر تقاوم الجفاف(ولكن لكل قاعدة في شواذ..ف في عدد قليل بقدر يقاوم الحرارة بس مو العالية كتير)

    - Conidiospores: external spores that are connected with hypha but not surrounded by a sack
    - Sporangiospores: spores that are inside a sack or a Sporangium (multilocular sporangia→ a sack with many rooms inside)
      Multilocular in particular means having many cells or compartments

  - Actinomycetes were firstly classified with fungi but they found out that they have the typical gram positive cell wall (which is the thick layer of peptidoglycan)

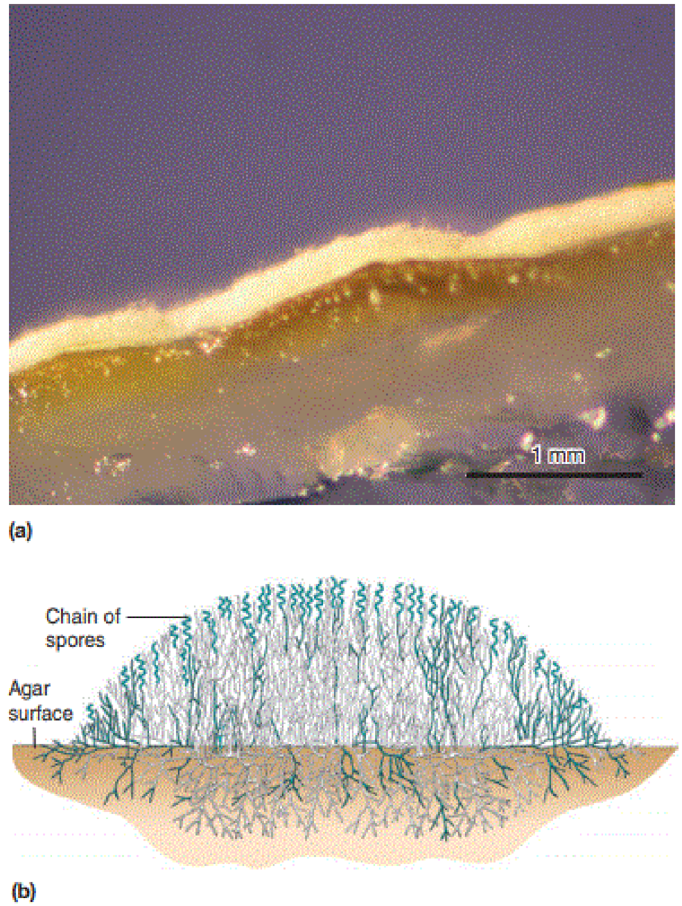


Figure 24.2 Cross Section of an Actinobacterial Colony. (a) This photomicrograph shows *Streptomyces griseus* substrate hyphae (yellow) growing into the agar and the white aerial hyphae growing away from the colony surface. (b) The colony consists of actively growing aerial hyphae that cannibalize the substrate hyphae to obtain nutrients. Live hyphae are blue; dead are white.

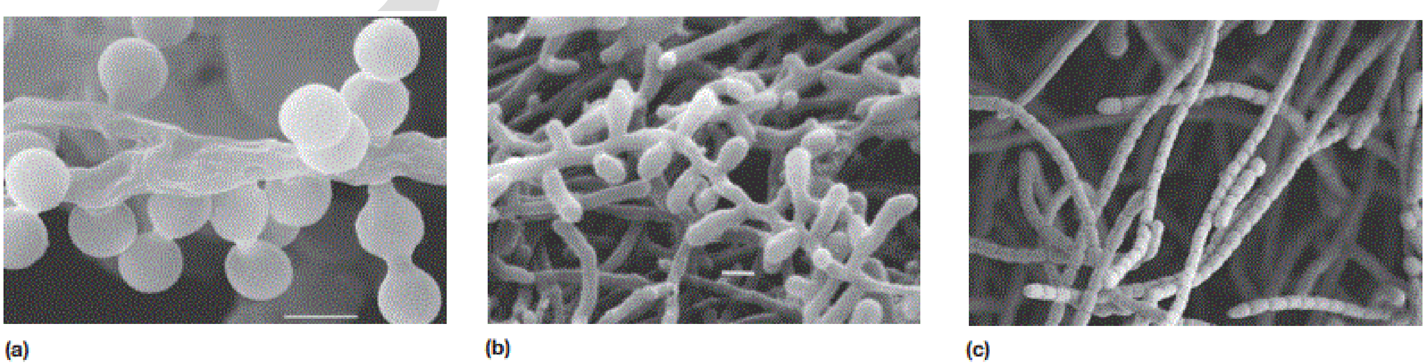


Figure 24.3 Examples of Actinomycete Spores as Seen in the Scanning Electron Microscope. (a) Spores of *Pseudonocardia yunnanensis* (bar = 1 μm). (b) Single spores of *Saccharomonospora viridis* are closely spaced along aerial hyphae. This organism causes farmer's lung. (c) Spore chain of *Streptomyces venezuelae*, which produces the antibiotic chloramphenicol.

- The classification of actinomycetes depends on:
  1. The production of diffusible pigments
  2. Type of cell wall(I,II,III,IV)→ differs in the structure of peptidoglycan
  3. Type of sugar present in the cell wall(A,B,C,D)
  4. The morphology and arrangement of spores

Table 24.2 Actinobacterial Cell Wall Types and Whole Cell Sugar Patterns

Cell Wall Type	Diaminopimelic Acid Isomer	Glycine in Interpeptide Bridge	Characteristic Sugars	Representative Genera
I	L, L	+	NA[1]	*Nocardioides, Streptomyces*
II	*meso*	+	NA	*Micromonospora, Pilimelia, Actinoplanes*
III	*meso*	−	NA	*Actinomadura, Frankia*
IV	*meso*	−	Arabinose, galactose	*Saccharomonospora, Nocardia*

Whole Cell Sugar Patterns[2]	Characteristic Sugars	Representative Genera
A	Arabinose, galactose	*Nocardia, Rhodococcus, Saccharomonospora*
B	Madurose[3]	*Actinomadura, Streptosporangium, Dermatophilus*
C	None	*Thermomonospora, Actinosynnema, Geodermatophilus*
D	Arabinose, xylose	*Micromonospora, Actinoplanes*

1 NA, either not applicable or no diagnostic sugars.
2 Characteristic sugar patterns are present only in wall types II–IV, those actinobacteria with *meso*-diaminopimelic acid.
3 Madurose is 3-*O*-methyl-D-galactose.

* cell extracts of actinomycetes with the four wall types also contain specific sugars useful for identification

➔ according to the Cell wall ..we have four cell wall types(**in Latin numbers**): they are distinguished according to the different features of peptidoglycan composition

➔ according to the different Sugar patterns (**using alphabet**) :each type has its own sugar

- other characteristics used in actinomycete taxonomy
  - morphology and color of the mycelium and sporangia
  - surface features and arrangement of spores
  - % GC in DNA
  - cell membrane phospholipid composition
  - heat resistance of spores

    → We're following the classical way of classification however, the newer (modern) techniques depends on applying <u>molecular analysis</u> for the actinomycetes taxonomy (using comparison of the 16S rRNA sequence)

- Ecological Significance of Actinomycetes
  - They are widely distributed in soil
  - They play important role in **mineralization** of organic matter
    بقدروا يحللوا مواد باقي البكتيريا ما بتقدر تحللها متل الشعر والاظافر
  - Most of them are free living, but a few are pathogens

Mineralization : degradation of more than 100 organic compounds in the soil

TABLE 24.3 Some Characteristics of Major Actinomycete Groups

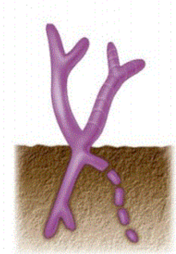
Group	Wall Type	Sugar Pattern	Mol% G + C	Spore Arrangement	Presence of Sporangia	Selected Genera
Nocardioform actinomycetes[a]	I, IV, VI[b]	A	59–79	Varies	–	*Nocardia, Rhodococcus, Nocardioides, Faenia, Oerskovia, Saccharomonospora*
Actinomycetes with multilocular sporangia	III	B, C, D	57–75	Clusters of spores	+(–)[c]	*Geodermatophilus, Dermatophilus, Frankia*
Actinoplanetes	II	D	71–73	Varies	Usually +	*Actinoplanes, Pilimelia, Dactylosporangium, Micromonospora*
Streptomyces and related genera	I	Not of taxonomic value	69–78	Chains of 5 to more than 50 spores	–	*Streptomyces, Streptoverticillium, Sporichthya*
Maduromycetes	III	B, C	64–74	Varies	+ or –	*Actinomadura, Microbispora, Planomonospora, Streptosporangium*
Thermomonospora and related genera	III	C (sometimes B)	64–73	Varies	–	*Thermomonospora, Actinosynnema, Nocardiopsis*
Thermoactinomycetes	III	C	52–55	Single, heat-resistant endospores	–	*Thermoactinomyces*

[a]Several genera have mycolic acid. Filaments readily fragment into rods and coccoid elements.
[b]Type VI cell wall has lysine instead of diaminopimelic acid.
[c]Members have clusters of spores that may not always be surrounded by a sporangial wall.

هلأ رح نحكي عن اهم الgroups بالactinomycetes :

A. Nocardioform actinomycetes :

- Genus Nocardia
  - Nocardia & Rhodococcus make up the family Nocardiaceae
  - develop a <u>substrate mycelium</u> that readily breaks into rods and coccoid elements
  - some also form an <u>aerial mycelium</u>



Nocardia

Impact of Nocardia :

- most are free-living saprophytes that can degrade many molecules:
  → petroleum hydrocarbons, detergents, benzene
  → involved in biodegradation of rubber joints in water and sewage pipes

- some are opportunistic pathogens causing **nocardiosis**: usually infect lungs and can also infect the central nervous system
- Nocardia & Rhodococcus are Important in mineralization

B. Actinoplanetes

- Suborder Micromonosporineae
  - extensive substrate mycelia
  - lack or have rudimentary (بواقي او بدائي)aerial mycelia
  - sporangiospores motile or nonmotile
  - found in soil and aquatic habitats (especially freshwater)

    → soil dwellers play important roles in plant and animal decomposition

    → some produce antibiotics like gentamicin

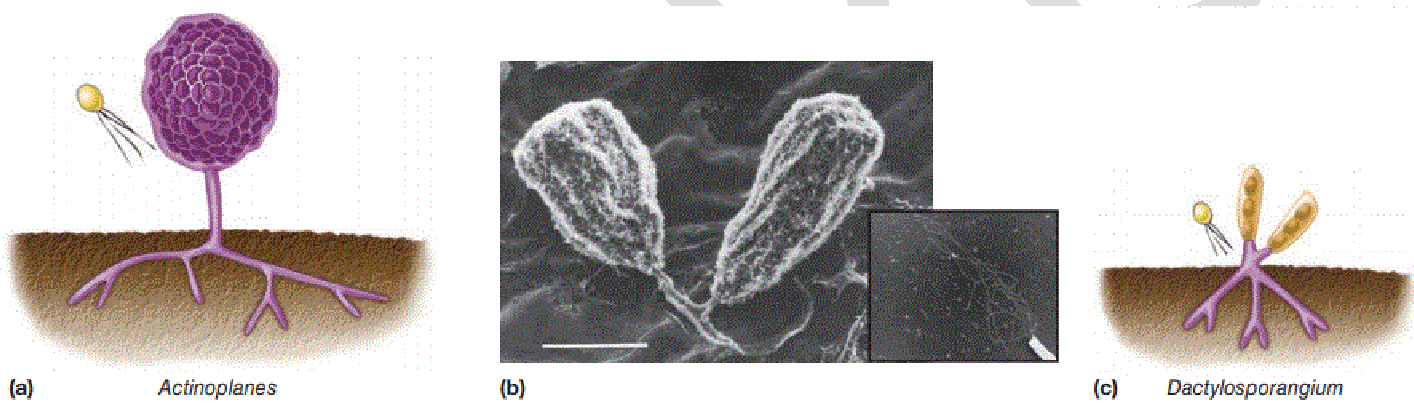


Figure 24.12 ***Micromonosporaceae.*** Actinoplanete morphology. (a) *Actinoplanes* structure. (b) A scanning electron micrograph of mature *Actinoplanes capillaceus* sporangia; inset shows a motile spore. (c) *Dactylosporangium* structure.

C. Streptomyces

- Streptomyces
  - The largest group of actinomycetes
  - are 1 to 20% of culturable soil microbiota
  - ➔ They produce **geosmin→ volatile substance that is source of moist earth odor**

    الgeosmin هو المادة الي منشم ريحتها لما تشتي الدنيا (ريحة الارض)
  - ➔ important in mineralization process

    can aerobically degrade many resistant substances(e.g., pectin lignin, and chitin)

    الchitin موجود بالجدار تبع الفطريات وهاي البكتيريا عندها الchitinase enzyme الي بقدر يحطم الchitin. ف وحدة من التطبيقات على هاد الانزيم في الbiotechnology انهم بدخلوه على النبات وبالتالي النبات بصير مقاوم للفطريات
  - produce vast array of antibiotics (most of the antibiotics that we use are produced by this bacteria)
  - most are nonpathogenic saprophytes

- have aerial hyphae that divide in single plane to form chains of 3-50 nonmotile spores
- all have type I cell wall
- G+C DNA content is 69-78%
  → They have conidiospores(not surrounded by a sack)
- Two pathogenic species of Streptomyces :
  - *Sterptomyces somaliensi* (the only HUMAN pathogen)
    →Causes **actinomycetoma:** infection of subcutaneous tissues in humans that leads to swelling, abscesses, and bone destruction

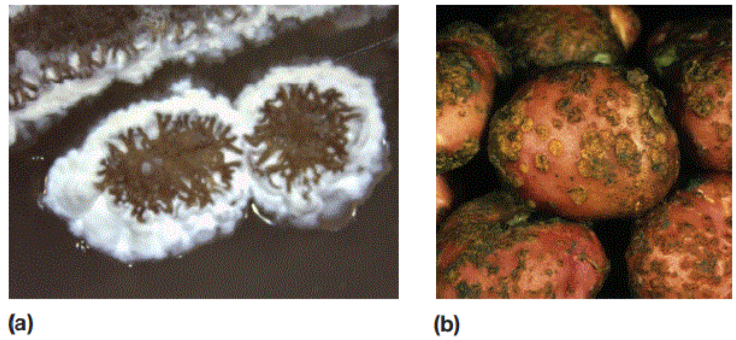


Figure 24.14 Streptomycetes of Practical Importance. (a) *Streptomyces griseus*. Colonies of the actinomycete that produces streptomycin. Each colony is about 0.5 cm in diameter. (b) *S. scabies* growing on a potato.

  - *Sterptomyces scabies*→ causes diseases in potato and beet(الشمندر)

D. Maduromycetes

- Suborder Streptosporangineae
  - aerial mycelia bear pairs or short chains of spores
    →whole cell homogenates contain sugar **madurose**(ومن هون اجى اسم الجروب)

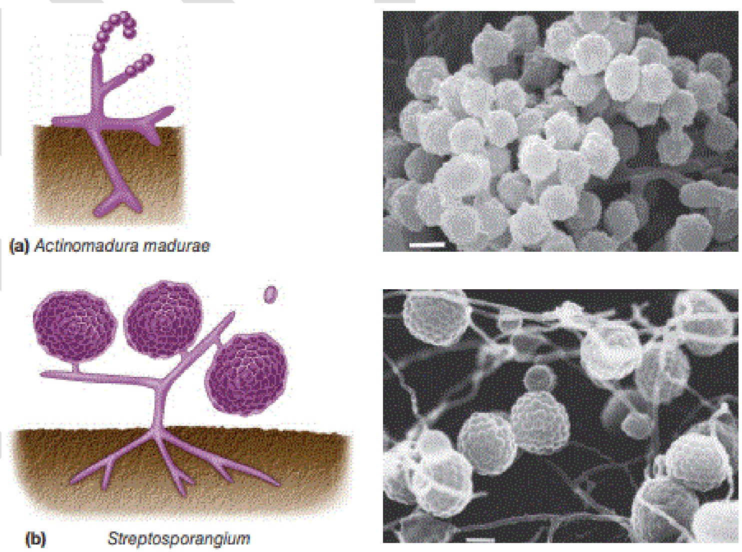
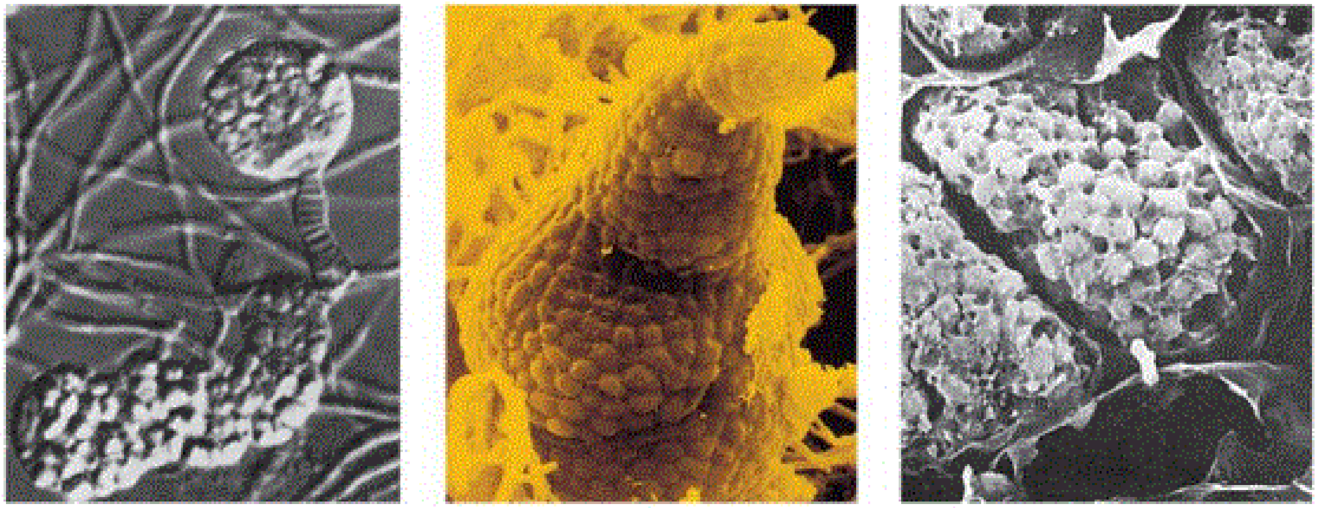


Figure 24.15 Maduromycetes. (a) *Actinomadura madurae* morphology; illustration and electron micrograph of a spore chain. (b) *Streptosporangium roseum* morphology; illustration and micrograph of sporangia and hyphae; SEM.

E. Actinomycetes with multilocular sporangia

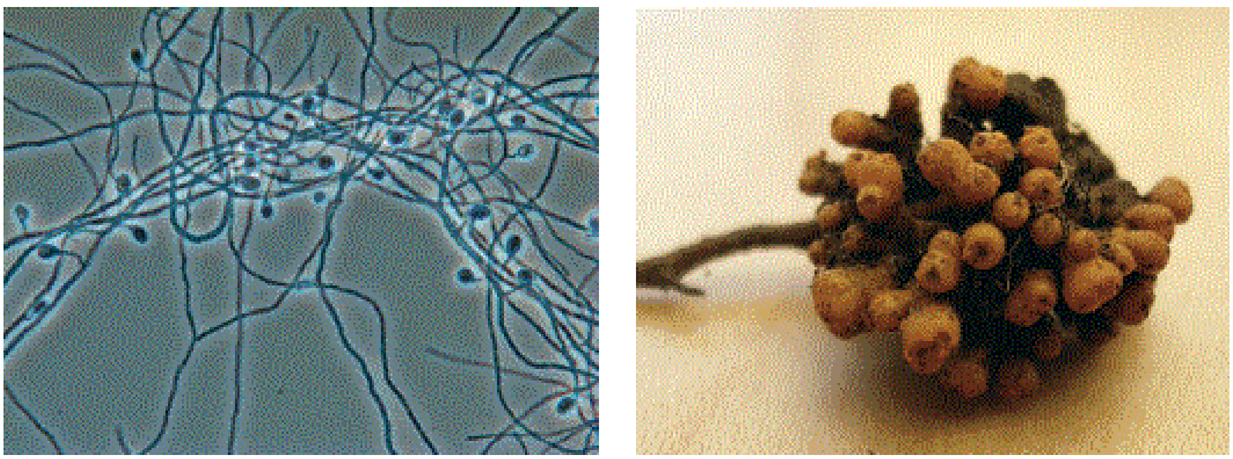
Multilocular يعني الspores مو بس موجودين بكيس ولكن هاد الكيس برضو مقسم لغرف (compartments)

- suborder Frankineae
  - contains the genera Frankia and Geodermatophilus
    → both form multilocular sporangia characterized by clusters of spores →both have type III cell walls
    - *Geodermatophilus*
      - motile spores
      - aerobic soil organisms
    - *Frankia*
      - non motile spores
      - microaerophile that can fix atmospheric nitrogen
        → the only actinomycetes that can fix atmospheric nitrogen But not freely in the soil; they do it with the association of trees in the forest (they can be found on the nodules of tree roots)



(a) (b) (c)

a: From S.T. Williams, M.E. Sharpe, and J.G. Holt (Eds.), *Bergey's Manual of Systematic Bacteriology*, Vol. 4, © 1989 Lippincott Williams and Wilkins Co., Baltimore; b: © R. Howard Berg/Visuals Unlimited; c: From S.T. Williams, M.E. Sharpe and J.G. Holt (Eds.), *Bergey's Manual of Systematic Bacteriology*, Vol. 4, © 1989 Williams and Wilkins Co., Baltimore



(a) **(b)**

Figure 24.16 ***Frankia.*** (a) A phase contrast micrograph showing hyphae, multilocular sporangia, and spores. (b) Nodules of the alder *Alnus rubra*.

F. Thermoactinomycetes

- Genus Thermoactinomyces
  - historically classified as actinomycete
  - more recently, phylogenetic analysis places it with low G+C microbes in order *Bacillales*, family *Thermoactinomycetaceae*
  - commonly found in high temperature environments such as composts

The End

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